

# **SIMATS ENGINEERING**

## **ASSESSMENT TOOL 3**

**COURSE CODE:** CSA51    **COURSE NAME:** Cryptography and Network Security

### **COURSE OUTCOME COVERED**

**CO3:** *Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation.CO (BL4 , BL5)*

Assessment Tool Weightage: 10%

**QUESTIONS: 5**

**TOTAL MARKS: 50**

### **Annexure A**

#### **MCQ**

#### ***Code of Conduct:***

*I \_\_\_\_\_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***

## **Annexure A**

### **MCQ**

#### **MCQ (Additional Set)**

1. Which property ensures the same input always produces the same hash?
  - A) Avalanche effect
  - B) Determinism
  - C) Collision resistance
  - D) Diffusion
2. What is the primary purpose of SHA-3?
  - A) Faster encryption
  - B) Alternative secure hashing standard
  - C) Data compression
  - D) Key exchange
3. Which attack tries all possible inputs to match a hash?
  - A) Replay attack
  - B) Brute-force attack
  - C) MITM attack
  - D) DoS attack
4. What is a salted hash?
  - A) Encrypted hash
  - B) Hash with random data added
  - C) Shortened hash
  - D) Reversed hash
5. Which algorithm is widely used for password hashing?
  - A) AES
  - B) RSA
  - C) bcrypt
  - D) DES
6. What does second pre-image resistance ensure?
  - A) Same input produces same output
  - B) Hard to find different input with same hash
  - C) Fast hashing
  - D) Small hash size
7. What is the output size of SHA-512?
  - A) 256 bits
  - B) 128 bits
  - C) 512 bits
  - D) 1024 bits
8. Which technique slows down brute-force attacks?
  - A) Encryption
  - B) Key exchange
  - C) Key stretching
  - D) Compression
9. What is the purpose of a nonce?
  - A) Encryption
  - B) Prevent replay attacks
  - C) Compress data
  - D) Generate keys
10. Which function is irreversible?
  - A) Encryption
  - B) Hashing

- C) Compression
  - D) Encoding
11. Which algorithm is used in DSS?
    - A) AES
    - B) DSA
    - C) DES
    - D) RC4
  12. What is required to verify a digital signature?
    - A) Private key
    - B) Public key
    - C) Session key
    - D) Shared key
  13. What does a digital signature guarantee?
    - A) Confidentiality
    - B) Integrity and authenticity
    - C) Compression
    - D) Speed
  14. What happens if message digest changes?
    - A) Signature remains valid
    - B) Verification fails
    - C) Encryption fails
    - D) Speed increases
  15. Which attack involves forging a signature?
    - A) Replay
    - B) Forgery
    - C) DoS
    - D) Sniffing
  16. What is the role of a hash in signatures?
    - A) Encrypt data
    - B) Reduce size
    - C) Ensure integrity
    - D) Store keys
  17. Which key is kept secret in signatures?
    - A) Public key
    - B) Private key
    - C) Session key
    - D) Shared key
  18. What ensures trust in certificates?
    - A) Encryption
    - B) Certificate Authority
    - C) Compression
    - D) Hash size
  19. What is certificate revocation?
    - A) Key generation
    - B) Canceling validity
    - C) Encryption
    - D) Compression
  20. What is OCSP used for?
    - A) Encryption
    - B) Certificate validation
    - C) Compression
    - D) Routing
  21. What is the main goal of Kerberos tickets?
    - A) Encryption
    - B) Authentication

- C) Compression
  - D) Routing
22. Which protocol uses tickets for authentication?
- A) SSL
  - B) Kerberos
  - C) FTP
  - D) HTTP
23. What is a session key?
- A) Permanent key
  - B) Temporary key
  - C) Public key
  - D) Private key
24. What is mutual authentication?
- A) One-way authentication
  - B) Both parties verify each other
  - C) Anonymous login
  - D) Weak authentication
25. What ensures freshness in Kerberos?
- A) Encryption
  - B) Timestamp
  - C) Compression
  - D) Hashing
26. What is a replay attack?
- A) Data modification
  - B) Resending valid data
  - C) Encryption
  - D) Compression
27. What is the function of an authenticator in Kerberos?
- A) Data storage
  - B) Identity proof
  - C) Encryption
  - D) Compression
28. What happens if system clocks differ in Kerberos?
- A) Faster authentication
  - B) Failure
  - C) Better security
  - D) No effect
29. What is the KDC composed of?
- A) Client & Server
  - B) AS & TGS
  - C) Firewall & IDS
  - D) Router & Switch
30. What is the AS in Kerberos?
- A) Application Server
  - B) Authentication Server
  - C) Access Server
  - D) Admin Server
31. What is X.509 certificate format used in?
- A) Networking
  - B) Digital certificates
  - C) Compression
  - D) Routing
32. What identifies the certificate owner?
- A) Public key
  - B) Subject field

- C) Signature
- D) Hash
- 33. What ensures certificate integrity?
  - A) Encryption
  - B) Digital signature
  - C) Compression
  - D) Routing
- 34. What is a root certificate?
  - A) Expired certificate
  - B) Self-signed trusted certificate
  - C) Invalid certificate
  - D) Temporary certificate
- 35. What is certificate pinning?
  - A) Storing keys
  - B) Binding certificate to host
  - C) Encryption
  - D) Compression
- 36. What is the risk of weak CA validation?
  - A) Faster speed
  - B) MITM attacks
  - C) Compression
  - D) Routing
- 37. What is the purpose of trust chain?
  - A) Encryption
  - B) Validate certificate hierarchy
  - C) Compress data
  - D) Routing
- 38. What is digital certificate used for?
  - A) Compression
  - B) Authentication
  - C) Routing
  - D) Storage
- 39. What ensures non-repudiation?
  - A) Encryption
  - B) Digital signature
  - C) Compression
  - D) Routing
- 40. What is a hash collision?
  - A) Same input
  - B) Same output for different inputs
  - C) Encryption
  - D) Compression
- 41. Which hash is strongest?
  - A) MD5
  - B) SHA-1
  - C) SHA-256
  - D) CRC
- 42. What is a one-way function?
  - A) Reversible
  - B) Irreversible
  - C) Fast
  - D) Small
- 43. What ensures authentication in PKI?
  - A) Encryption
  - B) Certificates

- C) Compression
  - D) Routing
44. What is key escrow?
- A) Key deletion
  - B) Key storage by third party
  - C) Encryption
  - D) Compression
45. What is identity spoofing?
- A) Encryption
  - B) Fake identity
  - C) Compression
  - D) Routing
46. What is trust anchor?
- A) Root CA
  - B) Client
  - C) Server
  - D) Router
47. What ensures integrity of message?
- A) Encryption
  - B) Hash
  - C) Compression
  - D) Routing
48. What is digital envelope?
- A) Hash
  - B) Encrypted message + key
  - C) Compression
  - D) Routing
49. What is certificate fingerprint?
- A) Encryption
  - B) Hash of certificate
  - C) Compression
  - D) Routing
50. What is PKI used for?
- A) Routing
  - B) Secure communication
  - C) Compression
  - D) Storage